# Kaushik Alaguvadivel Ramya

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## Research Interests

**Primary:** Self-Supervised Representation Learning, 3D/Volumetric Biomedical Imaging, Multiplexed Tissue Imaging, Computational Neuroscience, Vision Transformers, Graph Neural Networks

**Also:** Explainable AI, Fairness in Machine Learning, Biomedical Visual Analytics

## Education

**New York University** | *Master of Science in Computer Science* | GPA: 3.67/4.0 (as of Spring 2026) **May 2027**

- *Research Assistant:* NYU VIDA Center (from Spring 2026) – VIDA profile
- *Relevant Coursework:* Algorithms, Big Data, Information Visualization, Machine Learning, Responsible AI

**Thiagarajar College of Engineering** | *B.Tech in Information Technology* | GPA: 3.7/4.0 **May 2025**

- *Relevant Coursework:* Machine Learning, Artificial Intelligence, Data Mining, Data Analytics, Big Data Tools and Visualization, Algorithm Design, Fuzzy Sets and Clustering, Information Retrieval, Probability & Statistics

## Current Research

**MS Thesis: Self-Supervised 3D Learning for Multiplexed Tissue** **Jan 2026 – Present**

*NYU VIDA, Advisors: Prof. Robert Krueger and Prof. Erdem Varol (Krueger Lab) | 3D Self-Supervised Learning*

- Learning label-free, segmentation-free representations of the tumor microenvironment from 3D cyclic immunofluorescence (CyCIF) imaging of human melanoma, a terabyte-scale volumetric dataset with dozens of protein markers per voxel in which cells remain intact in 3D
- Pretraining 3D Vision Transformers with masked autoencoding (MAE) on full-resolution, all-channel volumes, and exploring DINO-style self-distillation as a complementary objective
- Studying how spatial context shapes what the model learns, comparing volumes that span part versus all of the tissue depth
- Building spatial graphs over learned region embeddings to model tissue neighborhoods with graph neural networks, without relying on 3D cell segmentation
- Unsupervised clustering of embeddings recovers spatially coherent tissue regions when mapped back onto the slide, despite no spatial information entering the clustering; training runs at scale on HPC with memory-efficient strategies

## Publications

**Enhanced Drone Classification using Transfer Learning and Optimized RF-Spectrogram** **2025** Link

- *Kaushik A R (first author)*
- *Presented at IEEE Conference, July 2025*
- Converted RF signals into optimized spectrogram images for drone classification; proposed a Cross Stage Partial Network with 5× lower computational complexity, 20× faster inference, and 5% accuracy improvement over nine CNN baselines.

**Enhancing Human Activity Recognition: An Exploration of Machine Learning Models and Explainable AI Approaches for Feature Contribution Analysis** **2024** Link

- *Kaushik A R (first author)*
- *ICEMCE 2023, Published February 21, 2024 — DOI: 10.1109/ICEMCE57940.2023.10434184*

- Applied ML models and LIME/SHAP to smartphone-sensor activity data from 30 participants; SHAP analysis of 500+ attributes reduced feature redundancy by 30% while maintaining 95%+ accuracy.

**Performance Analysis of Regression Models in Solar PV Forecasting** **2023** Link

- *Kaushik A R (first author)*
- *AISP 2023, Published June 1, 2023 — DOI: 10.1109/AISP57993.2023.10134943*
- Short-term solar PV power forecasting with Linear, Random Forest, and Decision Tree regression on irradiance, temperature, humidity, and wind data; ensemble achieved 99% accuracy for day-ahead prediction.

## Research Experience

**Summer Research, Computational Neuroscience** **June 2026 – July 2026**

*Neuroinformatics Lab, NYU, Advisor: Prof. Erdem Varol | Self-Supervised Learning, Neural Imaging*

- **Neuropixels spike sorting:** contributed to a self-supervised pipeline for spike localization and drift correction; audited multi-session recordings, helped resolve a cross-detector spike-matching issue, and built the training data framework for a spike-to-cluster classifier
- **Voltage imaging:** built a pipeline for sub-millisecond neural activity from low-SNR data, combining image reconstruction, motion correction, and BM4D denoising; tuned reconstruction kernels through a systematic sweep and scaled processing on HPC

**Machine Learning Research Assistant** **Sept 2022 – May 2025**

*Thiagarajar College of Engineering, Advisor: Dr. S. Padmavathi (spmcse@tce.edu) | Explainable AI, Fairness in ML, Deep Learning*

- First author on 3 published IEEE conference papers and 2 manuscripts in progress; 10+ citations on Google Scholar
- **Fair affect recognition** (submitted to *IEEE Transactions on Affective Computing*, code): GRU architecture for emotion classification from heterogeneous biosignals, more efficient than LSTM and Transformer baselines; evaluated DIR, Reweighing, and LFR across 11 fairness metrics (SPD, EOD, Theil Index); LFR cut bias by 30% at 95% accuracy
- **Explainable affect recognition** (in preparation, code): SHAP and LIME on multimodal physiological signals (ECG, EMG, GSR, RSP, BVP); identified GSR and RSP as primary arousal indicators, with ECG and GSR driving 60%+ of arousal decisions
- **Lightweight UAV threat detection** with the Defence Institute of Advanced Technology (DIAT), Pune (code): Cross Stage Partial Network with 5× lower complexity and 20× faster inference; 5% accuracy gain over SOTA across nine CNN architectures; models deployed

## Teaching

**Teaching Assistant, Biomedical Visual Analytics** **2026**

*New York University, Instructor: Prof. Robert Krueger | Biomedical Visualization*

**Teaching Assistant, Cloud Computing and Distributed Systems** **Aug 2023 – May 2025**

*Thiagarajar College of Engineering | 200+ students*

- Led lab sessions, graded assignments and exams, mentored students on AWS, Docker, and distributed architectures, and created supplementary course materials

## Technical Skills

- **Programming:** Python, MATLAB, Java, SQL
- **Deep Learning:** PyTorch, TensorFlow, Vision Transformers, Masked Autoencoders, DINO, Graph Neural Networks, CNNs, RNNs/GRUs/LSTMs, Transfer Learning
- **Large-Scale Training & HPC:** SLURM job arrays, mixed precision, gradient checkpointing, zarr/OME-Zarr, multi-GB/TB imaging data
- **Imaging & Signal Processing:** Image reconstruction, motion correction (Kalman-RTS), BM4D denoising, Neuropixels electrophysiology
- **ML Analysis & XAI:** Scikit-learn, SHAP, LIME, UMAP, clustering, fairness metrics and bias mitigation (DIR, Reweighing, LFR)
- **Tools:** Pandas, NumPy, SciPy, Git, Docker, AWS, Google Cloud, Jupyter, LaTeX

## Certifications

AWS Certified Machine Learning Engineer – Associate; AWS Certified AI Practitioner

## Honors & Awards

Selected for Demo-Day presentation for Energy Consumption Forecasting project at Thiagarajar College of Engineering

Teaching Assistant recognition for outstanding contributions to Cloud Computing and Distributed Systems courses